

Data analyst take-home

About 4.5 hours of work, 72 hours to send it back. Python only. We're reading your judgement, not your formatting.

Role	Company	Time	Tools
Data analyst	Tarang Pay (fictional)	~4.5 h · 72 h to submit	Python

Before you start

Tarang Pay is a payment gateway for Indian online businesses. Merchants use us to accept UPI, card, netbanking and wallet payments. We collect the money, keep a fee, and pay the rest out to the merchant.

Tarang Pay is invented for this assignment. There's nothing to look up. **Treat every rule below as true.**

We're preparing our investor update. Payment volume grew fast this year, but three teams don't agree on what the numbers mean:

- **Finance** says fee revenue in the books is lower than our volume should produce.
- **Operations** says merchants complain that some payments reach them late, or never.
- **The CFO** saw refunds spike in one week of August and wants to know if something broke.

Your job isn't to build a dashboard. It's to tell us **which numbers we can trust, where money is going missing, and what to fix first.**

If something is unclear, write the question down. Send a short list of the questions you would ask us, with the assumption you made for each. Asking a good question counts for you. Guessing silently counts against you.

Using AI tools is expected, not just tolerated. Task 7 asks you to show how you used them.

Rules of the business (treat as true)

Orders and attempts

- A customer places an order with a merchant. Each try to pay for it is an attempt, with its own `txn_id`.
- If an attempt fails, the customer may retry. That creates a new attempt with the same `order_id` and the next `attempt_no`.
- An order is successful if any of its attempts succeeded.

Payment status

- An attempt ends as success or failed. Until then it is pending, which must resolve within 48 hours.
- `status_updated_at` records when the final status reached Tarang Pay.

Settlement

- A successful payment is paid out to the merchant on the next working day (T+1). Working days exclude Sundays, the second and fourth Saturdays, and the bank holidays in `calendar.csv`.
- Day T is the day the payment's final success status reached us. A status that arrives at or after 23:00 IST counts toward the next day.
- Merchants whose KYC is `on_hold` are not paid out until it clears. That is correct behaviour, not an error.

Fees and GST

- Fee = payment amount × the rate in `pricing.csv` for that merchant and method, on the payment date.
- UPI carries a 0% fee for every merchant.
- GST at 18% is charged on the fee, not on the payment amount.
- The fee and GST are deducted at settlement. Finance books fee revenue (ex-GST) in `revenue_ledger.csv.gz`, one entry per settlement batch, on the settlement date.

Refunds

- A merchant can refund a successful payment, fully or partly, up to 90 days after it.

- Tarang Pay keeps its fee when a payment is refunded.

All money is in rupees. All times are IST. The data runs from 1 January to 31 August 2026.

The files

File	One row per	Rows
merchants.csv	merchant	2,010
pricing.csv	merchant × method × period a rate applies	11,840
transactions.csv.gz	payment attempt	506,062
settlements.csv.gz	payment paid out to a merchant	445,917
revenue_ledger.csv.gz	fee entry in finance's books (one per settlement batch)	72,876
refunds.csv	refund	13,013
calendar.csv	day, with a working-day flag and holiday name	243
monthly_report.csv	month, as currently reported to leadership	8

Columns

Column	Meaning
order_id, txn_id, attempt_no	Order, attempt, and which try it was (1 = first)
created_at	When the attempt started
status, status_updated_at	Final status, and when it reached us (blank if still pending)
failure_reason	For failed attempts: customer_dropped, insufficient_funds, bank_declined, bank_timeout, technical_error
method, bank	Payment method, and the customer's bank
settlement_id	Settlement batch: one per merchant per settlement date
payment_amount, fee_charged, gst_charged, settled_amount	The payment, what we kept, and what the merchant received
fee_booked, booked_date	Fee revenue as finance recorded it (ex-GST)
fee_rate_pct, valid_from, valid_to, plan	Agreed fee rate and when it applies (blank valid_to = still in force)
category, size_tier, joined_date	Merchant details
kyc_status, kyc_hold_since	KYC state, and the date a hold started

Task 1 • Trust the files • 60 min

Before analysing anything, check that the data means what it says.

- Profile each file: completeness, ranges, duplicates, impossible values.
- Then check across files. Does every settled payment have a successful attempt? Does every settlement batch have a ledger entry? Does every fee match the pricing table? Does every refund belong to a successful payment, and is any refund larger than its payment or later than 90 days?
- At least three things in these files are genuinely wrong, and at least one thing looks wrong but isn't.

For each finding, add one row to an investigation log:

What you noticed	Rows and ₹ affected	What you think happened	How you handled it	Why that handling
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"How you handled it" means one of: corrected, excluded, flagged but kept, or left alone. We care more about this choice than the name you give the problem. Changing financial data without proof is as bad as ignoring a real error. End with up to five questions you would ask an engineer or finance before trusting these numbers.

Task 2 • How the business is really doing • 55 min

- One row per month: attempts, orders, success rate, TPV (value of successful payments), average payment value, take rate (fee revenue ÷ TPV), refund rate, active merchants.
- You decide how to define success rate. Say which definition you used, and what the other one would have shown.

- Where do payments fail? By method, bank, hour of day and merchant category. Which failures could Tarang Pay fix, and which are the customer's choice?
- Compare your numbers with `monthly_report.csv`. Which reported numbers are wrong, by how much, and why?
- Take rate fell this year. Is that bad news? Two sentences.

The one trap we will warn you about. Money doesn't move on the day a payment happens. A settlement or a ledger entry on a given day belongs to payments from an earlier working day, and a refund belongs to a payment from days or weeks earlier. Comparing same-day totals across these files creates gaps that aren't real and hides some that are. Tell us which way your numbers would be wrong if you ignored this.

Task 3 · Where the money goes missing · 60 min

Follow successful payments through the chain: successful payment → paid out to the merchant → fee booked by finance. Report the settlement completion rate (successful payments that were paid out) and the on-time settlement rate (paid out on the next working day after the payment), by month. Then:

- The settlement gap. Some successful payments were paid out late or not at all. Split the gap into its causes. For each: rows, ₹, whether it is expected behaviour, a process problem or a data problem, and who should fix it.
- Fees. Compare the fee we should have charged (from the pricing table) with the fee we did charge. Where they differ: ₹, merchants, and when it started.
- Finance's books. Compare fees charged at settlement with fees booked. Is there a gap? Is it the same problem as the one above, or a different one?

Say "potential leakage" until the evidence proves money was lost, and tell us what evidence would prove it.

Task 4 · Merchants · 30 min

- Concentration. What share of TPV and of fee revenue comes from the top 10% of merchants?
- Retention. For merchants who joined each month from January to June, what share were still taking payments in each later month?
- The link. Do merchants whose payments often fail in their first month leave sooner? Show the evidence, and say what else could explain it.
- Finish with the ten merchants the account team should call this week, and one line on why for each.

Task 5 · One sprint · 30 min

Engineering can give payments operations one sprint in October. The candidates:

- A. Fix the integration with the bank that sends late status updates.
- B. Fix the pricing configuration, and recover the fees we didn't charge.
- C. Build a daily automated reconciliation that alerts when payments, settlements and the ledger don't match.

For each option, estimate the ₹ impact per month (or the risk it removes), with your assumptions and how confident you are. Pick one. Then give us:

- The evidence that would prove your choice wrong.
- One headline number for leadership. Say which number, and why that one.
- One metric for the monthly investor deck, defended in three sentences. We disagree internally between success rate, TPV and net take rate, so argue for yours rather than guessing what we want to hear.

Task 6 · One chart · 15 min

One chart, not a dashboard, that the Head of Payments could look at for ten seconds and know what to do. Underneath it, one sentence saying what it is telling them to do. If you can't write that sentence, it's the wrong chart.

Task 7 · How you used AI · 15 min

- Which tools you used, and for what: cleaning, joins, charts, writing.
- Your single best prompt, pasted exactly as you sent it.
- One thing the model got wrong, and how you caught it: a join that silently duplicated rows, a rate at the wrong grain, a confident claim the data doesn't support. "It was right about everything" reads as either no AI or no checking.

What to send back

- Your working: a Python notebook that reruns top to bottom from the files as we sent them.
- A one-page memo: recommendation first, evidence second, caveats last. One extra page is allowed for the investigation log.
- The chart, in the memo or as an image.
- Your questions list, with the assumption you made for each.

No machine learning. Nothing in these files needs a model, and reaching for one counts against you.

How we'll read it

We look for	Strong	Weak
Data judgement	The right handling for each problem, with the ₹ attached	Deletes rows silently, or corrects financial data without proof
Grain	Orders vs attempts, and payment date vs settlement date, used correctly	Joins that duplicate rows; same-day comparisons across files
Root cause	Splits the settlement gap into separate causes with owners	"There is a 3% gap"
Money	Every finding has a rupee amount	Percentages only
Decision	Picks one sprint and says what would change their mind	Recommends everything
Communication	A leader understands the memo without opening the notebook	The notebook is the answer